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ORIGINAL RESEARCH ARTICLE

Disproportionality Analysis and Characterisation of Medication Errors in EudraVigilance: Exploring Findings on Sexes and Age Groups

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Abstract

Background While medication errors (MEs) have been studied in the European Medicines Agency's EudraVigilance, extensive characterisation and signal detection based on sexes and age groups have not been attempted.

Objectives The aim of this study was to characterise all ME-related individual case safety reports in EudraVigilance and explore notable signals of disproportionate reporting (SDRs) among sexes and age groups for the 30 most frequently reported drugs.

Methods Individual case safety reports were used from EudraVigilance reported between 2002 and 2021. An ME was defined as any Preferred Term from the narrow Standardised Medical Dictionary for Regulatory Activities® Query. Signals of disproportionate reporting were selected based on a lower boundary of the 95% confidence interval ≥ 1 of the reporting odds ratio, and at least 3 individual case safety reports. Analysed subgroups were female individuals, male individuals, and age groups 0–1 month, 2 months to 2 years, 3–11 years, 12–17 years, 18–64 years, 65–85 years, and >85 years. Heatmaps were utilised as a visual aid to identify striking SDRs.

Results Of the 9,662,345 EudraVigilance reports, 267,262 (2.8%) contained at least one ME, with a total of 300,324 MEs, for 429,554 drugs. The most reported ME was "Inappropriate schedule of product administration" (52,646; 17.5%), followed by "Incorrect dose administered" (32,379; 10.8%) and "Wrong technique in product usage process" (26,831; 8.9%). Individual case safety reports with MEs were most frequently related to female individuals (148,009; 55.4%), most often submitted by healthcare professionals (155,711; 58.3%), originated predominantly from the USA (98,716; 36.9%), followed by France (26,678; 10.0%), and showed a median reported age of 50 years (interquartile range: 26–68). Most ME individual case safety reports (158,991; 59.5%) were associated with a serious health outcome. A total of 847 SDRs were identified, based on the entire EudraVigilance database; for subgroups, the number of SDRs ranged from 84 for the age group 0–1 month to 749 for female individuals. Signals of disproportionate reporting for female individuals and male individuals were very similar. Most MEs were reported for the vaccine against human papillomavirus (Anatomical Therapeutic Chemical [ATC]: J07BM01; 11,086 MEs, 57% being "inappropriate schedule of product administration"), with reporting odds ratios that range from 1.5 to 47.0 among age groups. The SDR for the live-attenuated vaccine against herpes zoster (ATC: J07BK02) had a reporting odds ratio that ranged from 26.6 to 78.1 among all subgroups. Signals of disproportionate reporting for oxycodone (ATC: N02AA05; 847 cases of "Accidental overdose", 35%), risperidone (ATC: N05AX08; 469 cases "Inappropriate schedule of product administration", 22.3%) and rivaroxaban (ATC: B01AF01; 1,377 cases of "Incorrect dose administered", 34.6%) stood out with higher magnitude SDRs for the age group 2 months to 2 years, with an reporting odds ratio range between 8.2 and 10.7, while for the entire EudraVigilance the reporting odds ratio ranged between 1.3 and 1.6 for the same drugs.

Conclusions This exploratory research provides an overview of characterised ME individual case safety reports and SDRs from the EudraVigilance database. Most conspicuous SDRs were identified in specific age groups. Signals of disproportionate reporting, not described in the literature, were found for vaccines, oxycodone, rivaroxaban and risperidone, and may prompt further examination by stakeholders. Top-reported MEs ("Inappropriate schedule of product administration", "Incorrect dose administered" and "Wrong technique in product usage process") emerged as a general priority focus to perform a further root-cause analysis involving healthcare providers, manufacturers and regulatory bodies, to improve the understanding and prevention of MEs.

Extended author information available on the last page of the article

Key Points

More than 800 drugs were found to be disproportionately reported in relation to medication errors.

Disproportionate reporting was different for various age groups, while it was similar for men and women.

Notable medication error findings, previously not described in the literature, were identified for vaccines, oxycodone, rivaroxaban and risperidone.

Combining the disproportionality analysis with the graphical aid of heatmaps provides a convenient method to explore, sort and visualise the abundance of results.

1 Introduction

A medication error (ME) is defined by the European Medicines Agency (EMA) as "an unintended failure in the drug treatment process that leads to, or has the potential to lead to harm to the patient" [1]. Previous research showed that MEs are increasingly reported over time [2, 3]. Increased reporting may be, not exclusively, attributed to the extended definition of adverse drug reactions introduced in European pharmacovigilance legislation in 2012 [4]. Despite known risk factors [5] and many preventative recommendations and guidelines [6–8], MEs persist worldwide up to this day [9–12].

While some MEs are harmless, other MEs may cause fatal outcomes such as death among various patient populations. It has been previously reported that 37% of individual case safety reports (ICSRs) on MEs were co-reported with a serious health outcome, of which 4% were co-reported with death [2]. A study from the USA found that in an outpatient setting, 15 ME-related adverse drug events per 1000 person-years were preventable [13].

In addition to health implications, MEs also impose a financial burden. For example, studies reported that the cost of an ME can be as low as €2.58 in Iraq, in the case of an invalid immunisation dose for a child [14], or as high as €6432 for in-hospital preventable adverse drug events in the Netherlands [15].

The prevalence of MEs in community settings ranged from 2% up to 94% [5]. This wide prevalence range is due to (i) variability in the studied types of MEs, (ii) different ME definitions, (iii) specific selection of study population, (iv) varied study settings and (v) use of various ME detection tools, while there is no established gold standard [5]. A meta-analysis on ME prevalence among paediatric inpatients reported that MEs occurred at a rate between 45 and 167 per 1000 patient days [16]. That same research found that prescribing errors were the most frequent of all studied MEs, with proportions between 4.1 and 58.1 per 100 orders [16]. Numerous factors contribute to MEs in the paediatric population such as errors in dose calculations based on age or body weight, lack of dosing information specifically for children, lack of the appropriate formulations or the inevitable use of off-label products [17].

Previous efforts have been made to characterise MEs from ICSRs submitted to EMA's EudraVigilance database including data up to the year 2015 [3]. However, there was no data published on other subjects of interest, such as patient characteristics, seriousness of ME ICSRs and signals of disproportionate reporting (SDRs), especially among subgroups based on sex and age. In the context of this exploratory study, it was of interest to observe whether conspicuous findings would emerge when performing analyses on subgroups based on sex and age. Recently, we described ME characteristics and SDRs using data from the Food and Drug Administration's Adverse Event Reporting System for MEs submitted up to the year 2020 [2]. However, the ME-related Food and Drug Administration's Adverse Event Reporting System data mostly (79%) consist of reports from the USA [2], while EMA's EudraVigilance has a higher representation for MEs reported from the European continent (29%) [3]. Exploring MEs in EudraVigilance will contribute to a broader perspective on MEs reported worldwide in addition to previously performed research. The objective of the current research was to provide an up-to-date characterisation of MEs reported in EudraVigilance and explore notable SDRs when performing subgroup analyses based on sex and age groups.

2 Methods

2.1 EudraVigilance and Data Access

The data source for this research was EMA's EudraVigilance database, which is a database that contains information from submitted ICSRs on adverse drug reactions, adverse drug events and MEs from sponsors of clinical trials, marketing authorisation holders and national competent authorities in the European Economic Area and beyond [3, 18, 19]. Different stakeholders, including regulatory authorities, academia, healthcare professionals and patients, can request data access to EudraVigilance according to EMA's access policy [20] or view the (limited) data that are publicly available [21]. The required data to perform analyses was supplied for ICSRs submitted to EudraVigilance from the 1 January, 2002 up

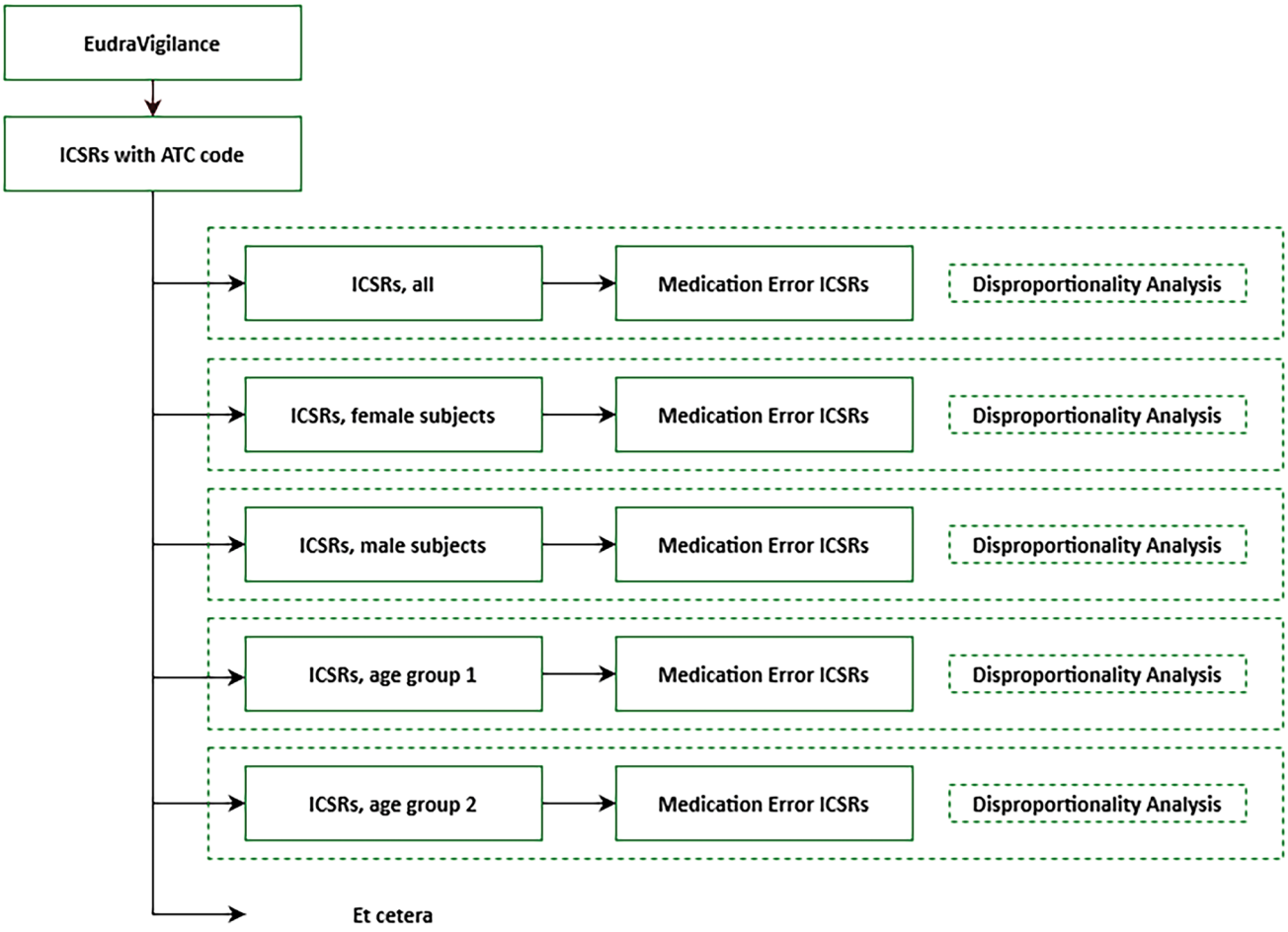

Fig. 1 Methodological overview on the selection of individual case safety reports (ICSRs) for analysis, including examples of analysis based on sexes and age groups. *ATC* Anatomical Therapeutic Chemical Classification

to 31 December, 2021. Only drugs that were marked by the EMA as "suspect" (the main drug suspected to be associated with the reported events) for the reported Preferred Term (PT) within the ICSRs were provided. The *Medical Dictionary for Regulatory Activities*® (MedDRA®) is used as a standardised medical terminology for the registration of a wide variety of events, including MEs, described in ICSRs.

2.2 Medication Errors

An ME was defined as a MedDRA® PT represented under the narrow Standardised MedDRA® Query (SMQ) for MEs, under MedDRA® version 24.1, published in September 2021. The narrow SMQ facilitates obtaining cases that are highly likely to represent MEs, focusing on specificity, which is in contrast with the broad SMQ that yields results with a higher sensitivity [22].

2.3 Data Processing and Analysis

An overview of the data processing and analysis is displayed in Fig. 1. The data were imported in a PostgreSQL database running on version 16.1. Aioli [23] is an extension of the Adverse Event Open Learning through Universal Standardization (AEOLUS) system [24], which was developed to standardise data from the Food and Drug Administration's Adverse Event Reporting System, but did not map drug names to Anatomical Therapeutic Chemical (ATC) classification system [25] codes. If for a given drug multiple ATC codes were available for different therapeutic use, the Aioli system determined the best fit using available information on drug strength and route of administration. If there was not sufficient information to distinguish between multiple ATC codes, or if no ATC code existed for the given substance, no ATC code was assigned.

Signals of disproportionate reporting were calculated with adapted SQL scripts from the AEOLUS system [24, 26]. Signals of disproportionate reporting were expressed

Table 1 Overview of report counts based on various drug-event combinations expressed by the letters A, B, C, and D

	Any PT from narrow ME SMQ	Other events
A drug of interest	A	B
All other drugs	C	D

ME medication error, *MedDRA*® Medical Dictionary for Regulatory Activities®, *PT* Preferred Term, *SMQ* Standardised MedDRA® Query

by calculating the reporting odds ratio (ROR) with a 95% confidence interval (CI). The following counts of drug-event combinations were used as input for the disproportionality analysis: (A) drug of interest – ME; (B) drug of interest - other event; (C) other drugs – ME; and (D) other drugs - other events (Table 1). From every unique ICSR, each registered PT from the narrow ME SMQ contributes to the ME counts as seen for counts A and C. The ROR and 95% CI were then calculated as follows:

$$\mathrm{ROR} = \frac{A/C}{B/D} \tag{1}$$

$$95\%\ \mathrm{CI} = e^{\ln(ROR) \pm 1.96\sqrt{\frac{1}{A}+\frac{1}{B}+\frac{1}{C}+\frac{1}{D}}}. \tag{2}$$

A signal was regarded as disproportionate if the case count was at least 3 and the lower boundary of the 95% CI was greater than 1, as stated by the EMA [27].

Multiple reporters and reporter countries may have been noted under a unique ICSR if the same case was reported by multiple reporters from possibly different countries. For the descriptive statistics on reporters and corresponding reporting country, the column "Primary Source for Regulatory Purposes" (column code: C.2.r.5) [28] was used to prevent a unique ICSR contributing multiple times.

2.4 Seriousness of Health Outcomes in ICSRs with MEs

The seriousness was described in the REACTION table provided by the EMA, by separate columns for "Death" (column code: E.i.3.2a), "Life threatening" (column code: E.i.3.2b), "Caused or prolonged hospitalisation" (column code: E.i.3.2c), "Disabling/ Incapacitating" (column code: E.i.3.2d), "Congenital Anomaly/Birth Defect" (column code: E.i.3.2e) and "Other medically important condition" (column code: E.i.3.2f). The last category is defined as "other situations that may not be immediately life threatening or result in death or hospitalisation but may jeopardise the patient or may require intervention to prevent one of the other outcomes listed (e.g. "Death", "Life threatening") [29]. Multiple columns for the seriousness may have been marked in an ICSR.

2.5 Processing and Displaying of SDRs

Because of a large output of results, SDRs were selected and displayed in the following way. A selection was made of 30 SDRs based on the most frequently reported drugs with co-occurring MEs within the entire EudraVigilance database. These SDRs were sorted based on descending RORs. If the criteria were met (case count at least 3 and lower boundary 95% CI greater than 1), corresponding SDRs for female individuals, male individuals, and age groups 0–1 month, 2 months to 2 years, 3–11 years, 12–17 years, 18–64 years, 65–85 years and >85 years, were displayed right next to the SDRs based on the total population.

The final results for SDRs and counts for various types of MEs are displayed in heatmaps. While heatmaps are mostly popular in the field of biology for displaying high-throughput data [30], they have also been shown to be useful for visualisation of disproportionality results [31]. The heatmaps represent values of interest (i.e. RORs and case counts) for the selected SDRs as previously described. Shades of colours, which range from white to red, represented values in which the brightest red cells represent the highest value, while lighter shades of yellow and white represent lower values. Interactive heatmaps with additional data are also provided as Electronic Supplementary Material (ESM). Heatmaps were made with the R package heatmaply, version 1.4.2 [32]. Alternatively, the top ten findings for the entire database, not restricted to EMA's case count or ROR criteria, were displayed by ordering based on (A) the most frequently reported drugs in association with MEs and (B) the drugs with the highest lower boundary 95% CI of the ROR, accompanied by the top three MEs for each drug.

3 Results

3.1 General Findings on ME ICSRs

Of the total of 9,662,345 ICSRs as provided by the EMA, there were 267,262 (2.8%) ICSRs containing information on at least one narrow ME, with a total of 300,324 reported MEs. The yearly proportion of ICSRs on MEs has been steadily increasing since 2004 (274 ICSRs; 1.1%) up to 2020 (27,549 ICSRs; 3.7%) as displayed in Fig. 2. In 2021, the number of ME-related ICSRs submitted to EudraVigilance (48,958 ICSRs) had almost doubled compared with 2020.

An overview of the most frequently reported MEs is displayed in Table 2. The five most frequently reported PTs for MEs account for more than half of all MEs, being "Inappropriate schedule of product administration" (52,646; 17.5%),

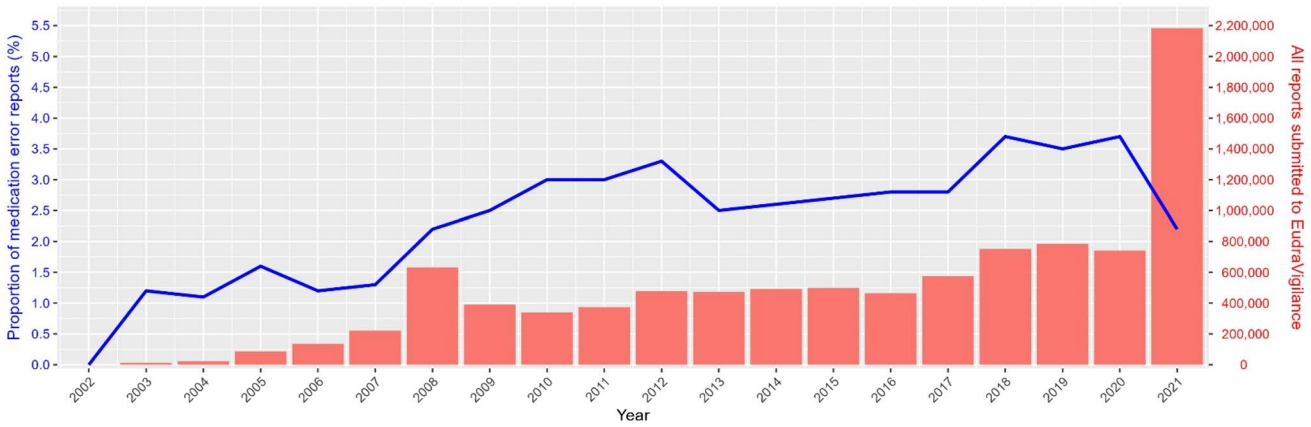

Fig. 2 Overview of all reports submitted to EudraVigilance (represented by the *red bar chart*, number of reports on the right *y-axis*) and the proportion of reports submitted mentioning a medication error (represented by the *blue line chart*, proportions noted on the left *y-axis*)

Table 2 Overview of the 15 most frequently reported medication errors in the EudraVigilance database

	Occurrence (%)
Total amount of reported medication errors	300,324 (100%)
Inappropriate schedule of product administration	52,646 (17.5%)
Incorrect dose administered	32,379 (10.8%)
Wrong technique in product usage process	26,831 (8.9%)
Medication error	21,394 (7.1%)
Product administration error	16,538 (5.5%)
Accidental overdose	15,011 (5%)
Incorrect route of product administration	14,017 (4.7%)
Incorrect product administration duration	10,733 (3.6%)
Product storage error	10,197 (3.4%)
Expired product administered	9137 (3%)
Accidental exposure to product	8452 (2.8%)
Product prescribing error	8208 (2.7%)
Contraindicated product administered	8059 (2.7%)
Wrong product administered	7114 (2.4%)
Product dispensing error	6066 (2%)

Table 3 Count overview of suspect drugs reported in reports on MEs

Analysed group	ICSRs on MEs/total (proportion)
Entire EudraVigilance	267,262/9,662,345 (2.8%)
Female	148,009/5,416,701 (2.7%)
Male	94,436/3,649,560 (2.6%)
0–1 month	1766/35,532 (5.0%)
2 months to 2 years	12,694/208,763 (6.1%)
3–11 years	10,254/192,970 (5.3%)
12–17 years	7929/199,299 (4.0%)
18–64 years	94,979/4,355,919 (2.2%)
65–85 years	50,441/2,075,852 (2.4%)
Above 85 years	8891/292,903 (3.0%)
ICSRs on MEs	267,262(100%)
Suspect drugs per report, *n* (%)	
1	215,133 (80.5%)
2	30,145 (11.3%)
3	8468 (3.2%)
4	3588 (1.3%)
5 or more	9912 (3.7%)
Suspect drugs found, *n* (%)	429,554 (100%)
Suspect drugs mapped to ATC	353,852 (82.4%)

ATC Anatomical Therapeutic Chemical Classification, *ICSR* individual case safety report, *ME* medication error

"Incorrect dose administered" (32,379; 10.8%), "Wrong technique in product usage process" (26,831; 8.9%), "Medication error" (21,394; 7.1%) and "Product administration error" (16,538; 5.5%). Characteristics were separately provided for ICSRs noting "Inappropriate schedule of product administration" and "Incorrect dose administered" in Tables S1–S3 of the ESM.

The highest proportion of MEs was found in the age group 2 months to 2 years (12,694 ME ICSRs; 6.1% of 208,763 group total ICSRs), while the lowest proportion of MEs were found in the age group 18–64 years (94,979 ME ICSRs; 2.2% of 4,355,919 group total ICSRs), as seen in Table 3. The majority of ME-related ICSRs (215,133 ICSRs; 80.5%) had one drug marked as the main "suspect" within the corresponding ICSR (Table 3). The total amount of suspect drugs for MEs was 429,554, of which 353,852 (82.4%) could be mapped to an ATC code. A list of unmapped drugs within the entire EudraVigilance database is provided in Table S4 of the ESM.

3.2 Person Characteristics

The ICSRs on MEs concerned more frequently female individuals (148,009; 55.4%) than male individuals (94,436;

Table 4 EudraVigilance person characteristics of medication error-related reports

	Count (proportion)
ICSRs on medication errors	267,262 (100%)
Sex	
Female	148,009 (55.4%)
Male	94,436 (35.3%)
Not reported	24,817 (9.3%)
Age category	
0–1 month	1766 (0.7%)
2 months to 2 years	12,694 (4.7%)
3–11 years	10,254 (3.8%)
12–17 years	7929 (3.0%)
18–64 years	94,979 (35.5%)
65–85 years	50,441 (18.9%)
Above 85 years	8891 (3.3%)
Not reported	80,308 (30.0%)
Median age, in years (IQR)	50 (26–68)

ICSR individual case safety report, *IQR* interquartile range

35.3%), while sex was unknown in 24,817 (9.3%) ICSRs (Table 4). The median reported age was 50 years (interquartile range: 26–68 years) and the age category of 18–64 years (94,979 ICSRs; 35.5%) was most frequently reported within the ICSRs on MEs (Table 4). The age category was missing in 80,308 ICSRs (30.0%).

3.3 Reporter and Reporter Country Characteristics

Most ICSRs on MEs were reported by healthcare professionals accounting for 155,711 (58.3%) ICSRs (Table 5). Additionally, a considerable amount of 103,128 (38.6%) ICSRs were submitted by consumers and non-healthcare professionals. Most ICSRs (98,716; 36.9%) originated from the USA. Countries from the European Union that reported most MEs were France (26,678 ICSRs; 10.0%), Germany (25,117 ICSRs; 9.4%) and Spain, (5,706 ICSRs; 2.1%). Many other countries from outside of Europe also submitted a considerable amount of ME ICSRs, such as the UK, Canada, Brazil, Japan, Australia and China (Table 5).

3.4 Seriousness of Reported MEs

In 159,670 (59.7%) of the ICSRs, at least one category representing the seriousness of the reported ME was noted (Table 6). The most reported category was "Other medically important condition" for 108,557 (40.6%) ICSRs. The second most frequently reported category was "Hospitalisation" in 59,993 (22.4%) ICSRs. For 107,592 (40.3%) ICSRs, none of the seriousness categories was reported.

Table 5 EudraVigilance reporter and reporter country characteristics in ICSRs on medication errors

ICSRs on medication errors	267,262 (100%)
Reporter type	Count (proportion)
Consumer or other non-health professional	103,128 (38.6%)
Physician	64,810 (24.2%)
Other health professional	57,325 (21.4%)
Pharmacist	33,576 (12.6%)
Lawyer	7329 (2.7%)
Unknown	1094 (< 1%)
Reporter country	Count (proportion)
USA	98,716 (36.9%)
France	26,678 (10%)
Germany	25,117 (9.4%)
UK	21,394 (8%)
Canada	16,743 (6.3%)
Brazil	9885 (3.7%)
Japan	9176 (3.4%)
Spain	5706 (2.1%)
Italy	4581 (1.7%)
Australia	4442 (1.7%)
China	4315 (1.6%)
Sweden	4165 (1.6%)
Netherlands	2747 (1%)
161 countries contributing <1%	33,597 (12.6%)

ICSRs individual case safety reports

Table 6 Reported seriousness of the health outcomes for individual case safety reports including medication errors

	Count (proportion)
Report of serious health outcome	159,670 (100%)
Type of seriousness	
Caused or prolonged hospitalisation	59,993 (22.4%)
Death	12,245 (4.6%)
Life threatening	7857 (2.9%)
Disabling/incapacitating	6090 (2.3%)
Congenital anomaly/birth defect	127 (< 1%)
Other medically important condition, unspecified	108,557 (40.6%)

3.5 Identified SDRs and Subpopulation Differences

A total of 2221 drug-ME combinations were found, of which 847 (38.1%) met the criteria for disproportionality based on the entire EudraVigilance database (Table 7). The amount of SDRs was as low as 84 for the age group of 0-1 month and as high as 749 for female individuals. Most MEs were reported for the papillomavirus vaccine (human types 6, 11, 16, 18; ATC: J07BM01; 11,086 MEs), followed by

Table 7 Amount of identified signals meeting the disproportionality criteria for the entire EudraVigilance database and groups based on sexes and age groups

Analysed group	DMECs	SDRs meeting criteria (proportion)
Entire EudraVigilance	2221	847 (38.1%)
Female individuals	2048	749 (36.6%)
Male individuals	1938	676 (34.9%)
0–1 month	314	84 (26.8%)
1–2 months	980	401 (40.9%)
2–11 years	975	334 (34.3%)
12–18 years	815	165 (20.2%)
19–64 years	1923	636 (33.1%)
65–85 years	1627	517 (31.8%)
Above 85 years	976	250 (25.6%)

DMEC drug-medication error combination, *SDR* signal of disproportionate reporting

infliximab (ATC: L04AB02; 9,236 MEs) and varicella, live attenuated (ATC: J07BK01; 5,623 MEs) as seen in Table 8. Sex-based subgroup SDRs showed minor differences between the sexes and also when compared with SDRs based on the overall EudraVigilance database, as shown in the SDR heatmap (Fig. 3 and ESM). When further analysing the heatmap of SDRs, the most distinct result in terms of age was found for etonogestrel (ATC: G03AC08), which showed an ROR of 60.4 (95% CI 14.4–252.8; three cases) for the age group 65–85 years. Of the total of 4501 MEs identified for etonogestrel, the most commonly reported ME was "Incorrect product administration duration" (2639 cases; 59%) as seen in the heatmap represented by Fig. 4 (and ESM).

Differences among age subgroups were also observed for vaccination against human papillomavirus (human types 6, 11, 16, 18; ATC: J07BM01), showing an ROR as low as 1.5 (95% CI 1.3–1.7; 314 cases) for the age group 3–11 years, a higher ROR of 23.8 (95% CI 18.2–31.2; 78 cases) for age group 2 months to 2 years, and an even higher ROR of 47.0 (95% CI 19.2–115.2; 7 cases) for the age group 65–85 years. The human papillomavirus vaccine was also the most commonly reported drug associated with MEs, accounting for 11,086 reported MEs, of which 6417 (58%) were attributed to "Inappropriate schedule of product administration".

It was also notable that the SDR for the vaccine against herpes zoster (live attenuated; ATC: J07BK02) was stronger in all age groups, except age group 0–1 month, than for the entire EudraVigilance population. The ROR ranged between 45.9 and 79.1 among subgroups, while for the entire database the ROR was 36.4 (95% CI 35.2–37.7; 4,756 cases). For female individuals and male individuals, the ROR was lower, being respectively 26.6 (95%: 25.3–28.0; 1967 cases) and 30.4 (95% CI 28.3–32.6; 1028 cases). The most commonly reported ME was "Incorrect route of product administration" (1282 cases; 27.0%).

Analysis on age subgroups showed conspicuous SDRs for the age group 2 months to 2 years for the drugs oxycodone (ATC: N02AA05; ROR = 8.2; 95% CI 6.4–10.5; 73 cases), rivaroxaban (ATC: B01AF01; ROR = 10.7; 95% CI 4.7–24.1; 7 cases), and risperidone (ATC: N05AX08; ROR = 9.6; 95% CI 7.2–12.7; 57 cases). Very few SDRs for the most frequently reported drugs were found for the age group 0–1 months, but notably the SDR for paracetamol (ATC: N02BE01) stood out showing an ROR of 6.7 (95% CI 5.5–8.3; 100 cases). The most reported ME for oxycodone was "Accidental overdose" (847 cases; 35%), for risperidone "Inappropriate schedule of product administration" (469 cases; 22.3%), for rivaroxaban "Incorrect dose administered" (1377 cases; 34.6%) and for paracetamol "Accidental overdose" (1191 cases, 23%).

3.6 Type of MEs Among SDRs

Based on the results displayed in the heatmap in Fig. 4, several MEs stand out for a couple of drugs not yet mentioned above. For infliximab (ATC: L04AB02), there were 4707 out of a total of 9236 MEs (51%) attributed to "Inappropriate schedule of product administration". The ME "Incorrect dose administered" accounted for 4836 (89%) out of 5433 MEs for ranitidine (ATC: A02BA02). The ME "Contraindicated product administered" was reported in a range of 1785–3323 MEs for (in ascending order) golimumab (ATC: L04AB06; 60.1% of MEs), certolizumab pegol (ATC: L04AB05; 78.3%), sulfasalazine (ATC: A07EC01; 89.6%), infliximab (ATC: L04AB02; 23.7%), hydroxychloroquine (ATC: P01BA02; 88.1%), adalimumab (ATC: L04AB04; 53.3%), tocilizumab (ATC: L04AC07; 88.0%) and abatacept (ATC: L04AA24; 87.9%).

3.7 Frequent MEs Among Frequently Reported Drugs and Strongest SDRs

Tables 9 and 10 display the top ten findings, not restricted to the EMA's case count or ROR criteria, when ordering based on (A) the most frequently reported drugs in association with MEs and (B) the drugs with the highest lower boundary 95% CI of the ROR, accompanied by the top three MEs. A noteworthy result was the vaccine against coronavirus disease 2019 (ATC: J07BN01), which was the second most reported drug on MEs (10,150 ICSRs) in the database, predominantly because of the ME "Inappropriate schedule of product administration" (7510 cases; 74%). Another noteworthy result was etanercept (ATC: L04AB01) accounting for 8286 reported MEs, of which the most frequent ME was "Contraindicated product administered" (3135 cases; 37.8%). The drug with the highest lower boundary of the

Table 8 Thirty most reported drugs contributing to a signal of disproportionate reporting in relation to medication errors

Drug	ATC	Reported medication errors
Papillomavirus (human types 6, 11, 16, 18)	J07BM01	11,086
Infliximab	L04AB02	9236
Varicella, live attenuated	J07BK01	5623
Pneumococcus purified polysaccharides antigen and *Haemophilus influenzae*, conjugated	J07AL52	5534
Insulin glargine	A10AE04	5532
Ranitidine	A02BA02	5433
Paracetamol	N02BE01	5183
Adalimumab	L04AB04	4972
Zoster, live attenuated	J07BK02	4756
Etonogestrel	G03AC08	4501
Rivaroxaban	B01AF01	3983
Abatacept	L04AA24	3781
Insulin lispro	A10AB04	3623
Measles, combinations with mumps and rubella, live attenuated	J07BD52	3299
Tocilizumab	L04AC07	3136
Leflunomide	L04AK01	3061
Golimumab	L04AB06	2970
Hydroxychloroquine	P01BA02	2905
Insulin aspart	A10AB05	2881
Hepatitis b, purified antigen	J07BC01	2774
Pneumococcus, purified polysaccharides antigen conjugated	J07AL02	2685
Oxycodone	N02AA05	2419
Certolizumab pegol	L04AB05	2312
Sulfasalazine	A07EC01	2224
Acetylsalicylic acid	B01AC06	2124
Risperidone	N05AX08	2105
Nicotine	N07BA01	2084
Salbutamol	R03AC02	2081
Salmeterol and fluticasone	R03AK06	2053
Rota virus, pentavalent, live, reassorted	J07BH02	1969

ATC Anatomical Therapeutic Chemical Classification

95% CI of the ROR was sodium lauryl sulfoacetate, including combinations (ATC: A06AG11; 17 MEs; ROR with 95% CI 95, 48–189), with "Accidental device ingestion" (ten cases; 58.8%) being the most reported ME.

4 Discussion

4.1 SDRs and MEs

In this paper, we extensively characterised ME ICSRs, as these were underexposed in previous research [3], and explored SDRs for MEs, demonstrating notable differences in SDRs when stratifying by age. A reason why more differences were found between age groups than between sexes might be because of age being a more prominent factor than sex in altering medication needs for different age groups, for example by age-related dosing regimes, age-related diseases, increased polypharmacy over time, and more complex healthcare for the youngest and oldest populations. Heatmaps proved to be a convenient graphical tool to give a quick overview of SDRs on MEs, and compare the SDRs for different sexes and age groups.

4.2 Disproportionality Analysis as a Prioritisation Method

We showed that there are multiple ways to organise, sort and display the data, which brings both challenges and opportunities to discover new insights. The disproportionality analysis may be viewed in the context of this research as a prioritisation tool when dealing with an abundance of

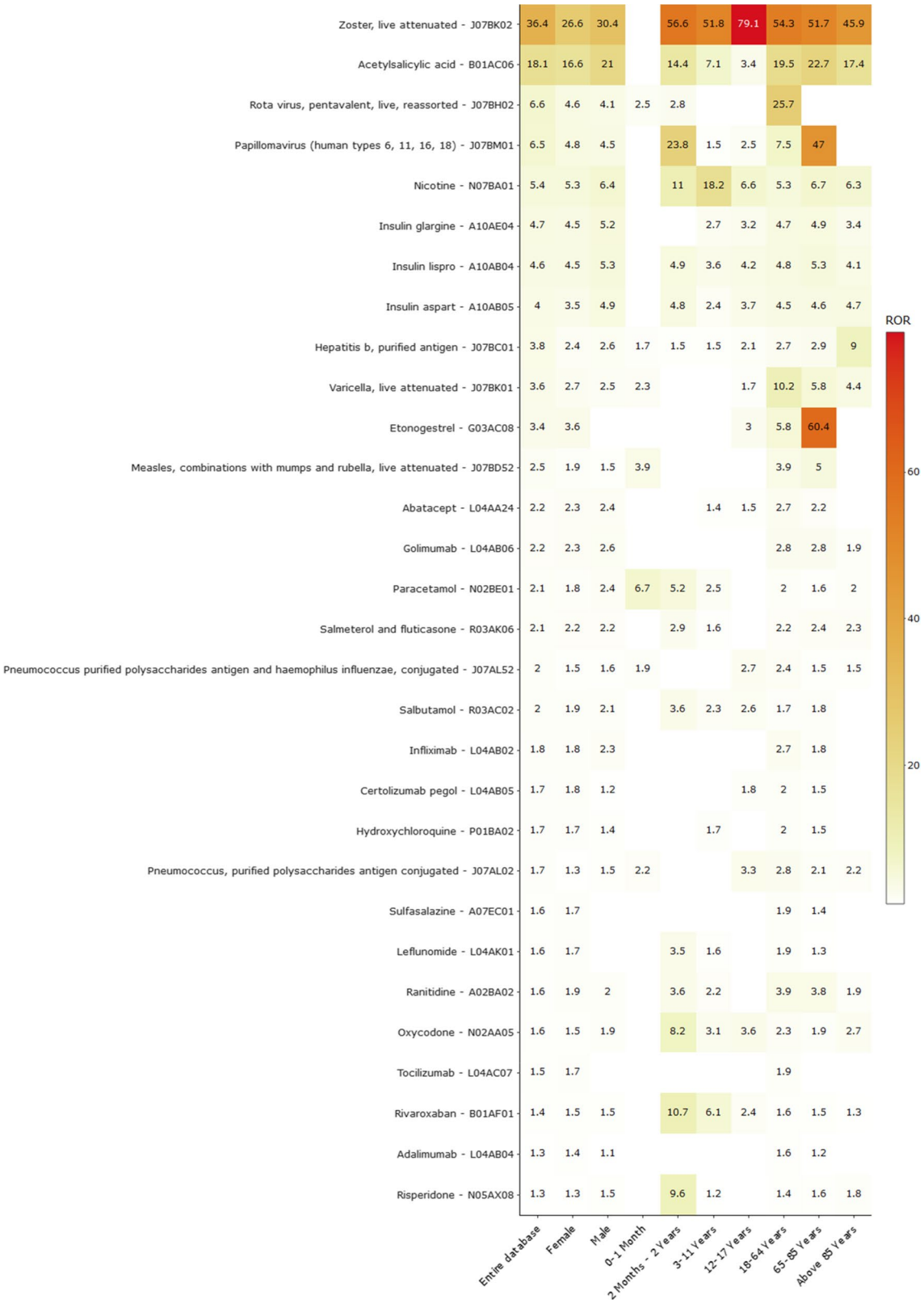

Fig. 3 Thirty most commonly reported drugs with their corresponding code from the Anatomical Therapeutic Chemical Classification System, associated with medication errors, ordered by the magnitude of disproportionality based on the values obtained when analysing the entire EudraVigilance database. Values of the reporting odds ratio are displayed where available for all the subgroups based on sexes and age groups. An *empty cell* signifies either no medication errors found or disproportionality criteria were not met. Interactive figure available in the ESM

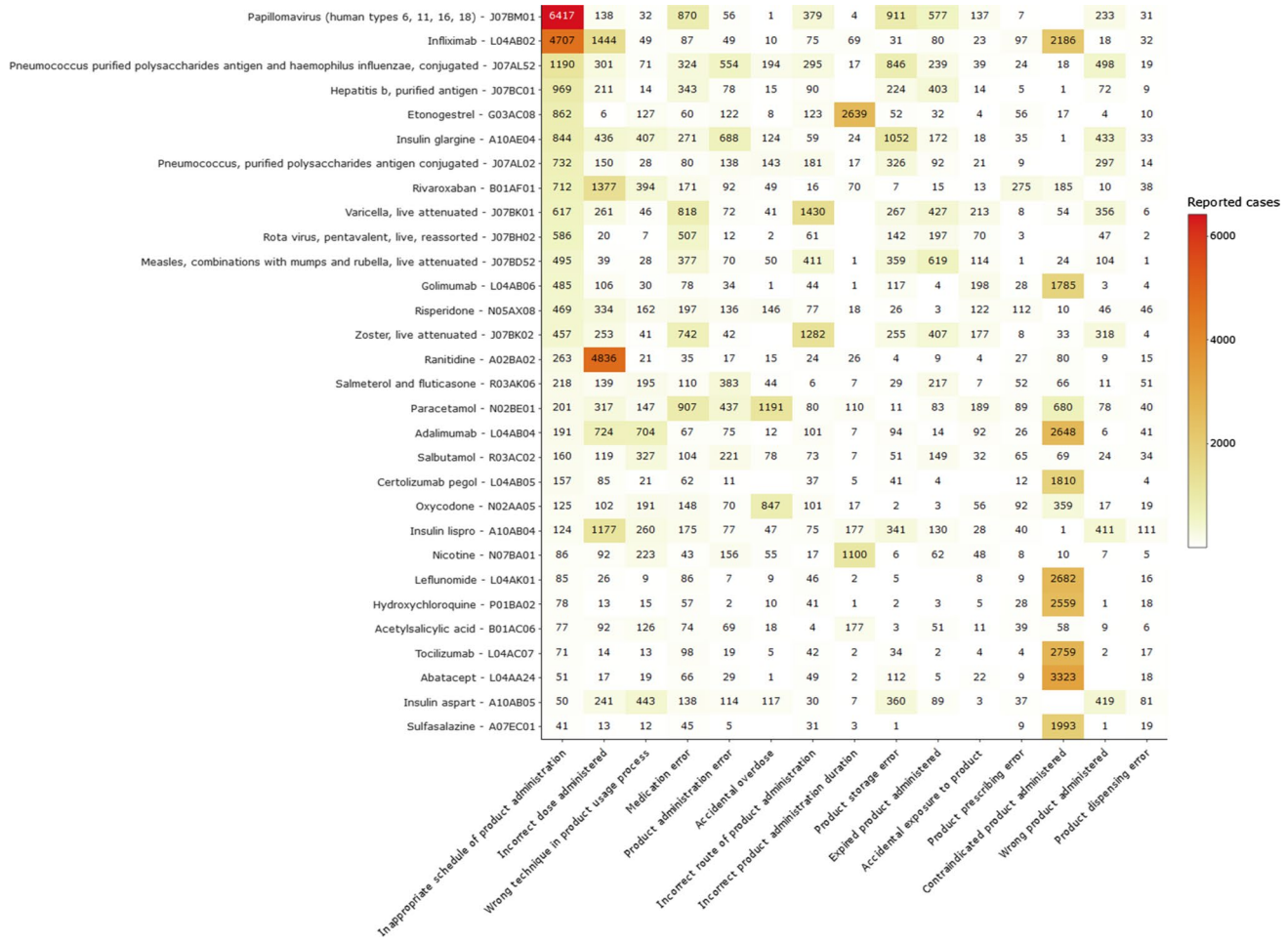

Fig. 4 Fifteen most commonly reported medication errors highlighted and quantified for the 30 most commonly reported drugs associated with medication errors based on the entire EudraVigilance database. Values of the amount of reported cases are displayed where available. An *empty cell* signifies that the corresponding type of medication error was not reported for a drug. Interactive figure available in the ESM

data. While sorting solely based on counts provides insight into the most frequently reported MEs, the disproportionality analysis provides more context given all the drugs and PTs reported in the EudraVigilance database. The disproportionality measure provides insight in drug-ME combinations that are more often reported than expected, given the data in EudraVigilance, which is a feature not present when solely sorting by counts. Sorting additionally on only counts showed how the coronavirus disease 2019 vaccine (ATC: J07BN01) was the second most reported drug on MEs; however, the low ROR (and narrow 95% CIs) could indicate that other PTs not related to MEs might have been disproportionately reported for the coronavirus disease 2019 vaccine. Sorting based on the lower boundary 95% CI highlighted other drugs for which MEs were reported disproportionally, but simultaneously showed that most findings were subject to a low case count, questioning the impact of these findings. By highlighting these different methodologies, we showed how different results may be obtained. These complementary results show the value of utilising different methodologies, and in particular the disproportionality analysis.

4.3 SDRs Previously Described in the Literature

Observations based on the heatmap revealed findings that relate to previous publications on MEs for specific drugs, such as various types of insulin (ATC: A10AE04, A10AB04, A10A05) [33], rivaroxaban (ATC: B01AF01) [34] and oxycodone (ATC: N02AA05) [35, 36]. Nicotine replacement therapy (ATC: N07BA01) is also known to be frequently reported in the context of MEs [37]; however, misuse might play a major role when considering disproportionality rates that were found among individuals aged <18 years in our study. While the varicella zoster vaccine live (Zostavax®, ATC: J07BK02) was only approved for people aged 50 years and above [38], we identified

Table 9 Ten most reported drugs on MEs with the three most commonly reported ME type per drug, omitting the Preferred Term "Medication Error"

ATC	Drug name	ROR (95% CI)	MEs reported for drug	ME	Amount reported
J07BM01	Papillomavirus (human types 6, 11, 16, 18)	6.51 (6.39–6.64)	11,086 (2.8%)	Inappropriate schedule of product administration	6417 (57.9%)
				Product storage error	911 (8.2%)
				Expired product administered	577 (5.2%)
J07BN01	COVID-19, RNA-based vaccine	0.28 (0.28–0.29)	10,150 (2.5%)	Inappropriate schedule of product administration	7510 (74%)
				Incorrect route of product administration	412 (4.1%)
				Expired product administered	384 (3.8%)
L04AB02	Infliximab	1.84 (1.81–1.88)	9236 (2.3%)	Inappropriate schedule of product administration	4707 (51%)
				Contraindicated product administered	2186 (23.7%)
				Incorrect dose administered	1444 (15.6%)
L04AB01	Etanercept	0.87 (0.85–0.88)	8286 (2.1%)	Contraindicated product administered	3135 (37.8%)
				Wrong technique in product usage process	1181 (14.3%)
				Product storage error	648 (7.8%)
J07BK01	Varicella, live attenuated	3.60 (3.50–3.69)	5623 (1.4%)	Incorrect route of product administration	1430 (25.4%)
				Inappropriate schedule of product administration	617 (11%)
				Extra dose administered	441 (7.8%)
J07AL52	Pneumococcus purified polysaccharides antigen and *Haemophilus influenzae*, conjugated	2.02 (1.97–2.08)	5534 (1.4%)	Inappropriate schedule of product administration	1190 (21.5%)
				Product storage error	846 (15.3%)
				Product administration error	554 (10%)
A10AE04	Insulin glargine	4.70 (4.57–4.83)	5532 (1.4%)	Product storage error	1052 (19%)
				Inappropriate schedule of product administration	844 (15.3%)
				Product administration error	688 (12.4%)
A02BA02	Ranitidine	1.56 (1.51–1.60)	5433 (1.4%)	Incorrect dose administered	4836 (89%)
				Inappropriate schedule of product administration	263 (4.8%)
				Contraindicated product administered	80 (1.5%)
N02BE01	Paracetamol	2.06 (2.00– 2.11)	5183 (1.3%)	Accidental overdose	1191 (23%)
				Contraindicated product administered	680 (13.1%)
				Product administration error	437 (8.4%)
L04AB04	Adalimumab	1.28 (1.25–1.32)	4972 (1.2%)	Contraindicated product administered	2648 (53.3%)
				Incorrect dose administered	724 (14.6%)
				Wrong technique in product usage process	704 (14.2%)

ATC Anatomical Therapeutic Chemical Classification, *CI* confidence interval, *COVID-19* coronavirus disease 2019, *ME* medication error, *RNA* ribonucleic acid, *ROR* reporting odds ratio

Table 10 Drugs related to MEs ordered by descending lower boundary 95% CI of the ROR with the three most commonly reported ME type per drug, omitting the preferred term "Medication Error"

ATC	Drug name	ROR (95% CI)	MEs reported for drug	ME	Amount reported
A06AG11	Sodium lauryl sulfoacetate, including combinations	95 (48–189)	17 (0.0046%)	Accidental device ingestion	10 (58.8%)
				Wrong technique in device usage process	4 (23.5%)
				Expired device used	2 (11.8%)
S01AE03	Ciprofloxacin	157 (46–537)	7 (0.0019%)	Incorrect route of product administration	3 (42.9%)
				Product dispensing error	3 (42.9%)
				Product administration error	1 (14.3%)
S01AD03	Aciclovir	97 (44–213)	8 (0.0021%)	Product dispensing error	4 (50%)
				Product administration error	2 (25%)
				Incorrect route of product administration	1 (12.5%)
S01AE01	Ofloxacin	53 (38–74)	41 (0.011%)	Incorrect route of product administration	12 (29.3%)
				Wrong product administered	11 (26.8%)
				Product dispensing error	5 (12.2%)
A01AB18	Clotrimazole	66 (35–124)	16 (0.0043%)	Incorrect route of product administration	11 (68.8%)
				Wrong product administered	1 (6.2%)
				Expired product administered	1 (6.2%)
J07BK02	Zoster, live attenuated	36 (35–38)	4,014 (1.1%)	Incorrect route of product administration	1,282 (31.9%)
				Inappropriate schedule of product administration	457 (11.4%)
				Expired product administered	407 (10.1%)
S01AX18	Povidone-iodine	55 (34–87)	13 (0.0035%)	Incorrect route of product administration	5 (38.5%)
				Accidental exposure to product	4 (30.8%)
				Wrong product administered	2 (15.4%)
S02AA01	Chloramphenicol	90 (34–239)	8 (0.0021%)	Product dispensing error	2 (25%)
				Product administration error	1 (12.5%)
				Product storage error	1 (12.5%)
D08AG02	Povidone-iodine	49 (31–78)	28 (0.0075%)	Product administration error	8 (28.6%)
				Expired product administered	7 (25%)
				Wrong technique in product usage process	5 (17.9%)
R05DA01	Ethylmorphine	40 (31–52)	78 (0.021%)	Wrong dose	77 (98.7%)
				Wrong product administered	1 (1.3%)

ATC Anatomical Therapeutic Chemical Classification, *CI* confidence interval, *ME* medication error, *ROR* reporting odds ratio

SDRs for infants, children and adolescents. In 2015, the New Zealand's Medicines and Medical Devices Safety Authority (Medsafe) received five reports on MEs related to the varicella zoster virus vaccines, of which four reports indicated that Zostavax® was erroneously administered to children [39]. The Vaccine Adverse Event Reporting System data, consulted via the Centers for Disease Control's Wonder search engine, showed 140 reports submitted on Zostavax® being administered to patients of inappropriate age [40]. A possible reason for these MEs as reported in the literature might be a mix-up with other vaccines that are stored together such as Varivax®, Varilrix®, Shingrix® and Zostavax® [39, 41, 42].

4.4 Known SDRs with an Unknown Root Cause

Other findings based on the heatmap show SDRs previously described in the literature, however, without a described root cause for MEs. Based on Vaccine Adverse Event Reporting System data, it has been found that the human papillomavirus vaccines have been administered to children of an inappropriate age, namely 0–8 years [43], corresponding to our findings as well. However, it is unclear how this may have happened. Similarly, it appears that the pentavalent rotavirus vaccine has been given to individuals in the inappropriate age category 18-64 years, generating an SDR for MEs. Previous literature has described how this vaccine was administered to children of inappropriate age [44]; however,

it is unclear how vaccination has occurred among an adult population not to be expected to receive this vaccine. It has also been previously reported that between 2001 and 2012, 160 ICSRs were submitted to the French poison centres on MEs among children exposed to risperidone (ATC: N05AX08) [45]. Most children (112) received an incorrect dose, of which 20 children were aged younger than 5 years, while risperidone was wrongly administered to 21 children [45]. From that publication, it is not clear how many children were found in the age category 2 months to 2 years, but the median reported age was 8 years with a range of 0.9–12 years. Risperidone was Food and Drug Administration approved for children aged 5 years and above [46], and it is concerning to see a disproportionate reporting among the age group 2 months to 2 years.

4.5 New SDRs, Not Known in the Literature

Based on the heatmaps, some previously unknown SDRs on MEs were also found. Notably, six drugs (i.e. infliximab [ATC: L04AB02], adalimumab [ATC: L04AB04], golimumab [ATC: L04AB06], certolizumab [ATC: L04AB05], tocilizumab [ATC: L04AC07] and abatacept [ATC: L04AA24]), classified as disease-modifying antirheumatic drugs (DMARDs) and more specifically biological DMARDs, were disproportionately reported, with the ME "Contraindicated product administered" being most frequent. While so far no research has been published on MEs for these drugs, it is known that biological DMARDs increase the risk of infections for individuals with rheumatoid arthritis [47–50], and guidelines state that biological DMARDs should not be prescribed in the presence of a serious active infection [48]. Similar findings were identified for SDRs on the DMARDs sulfasalazine (ATC: A07EC01) and hydroxychloroquine (ATC: P01BA02). These DMARDs are known to be used for the treatment of rheumatoid arthritis as part of complex guidelines [51], which might make these drugs more prone to MEs.

4.6 Strengths

This is the first study that highlighted the importance of a subgroup analysis, especially based on age, when examining SDRs for MEs within the EudraVigilance database. Differences between age groups may motivate further investigation of findings for specific age groups. The graphical representation of SDRs and providing an interactive version to further explore the data offer the possibility to get a clear overview of the obtained results, visualise notable results and explore undiscussed SDRs. The disproportionality analysis may be seen as a method to sort results and prioritise further investigation of SDRs previously not known in literature. It might trigger stakeholders to further examine these findings. This exploratory study showed several of many possibilities to sort the findings, and could inspire follow-up research on the ME topic or specific ME findings.

4.7 Limitations

A well-known inherent limitation of all spontaneous reporting systems, including EudraVigilance, is under-reporting [52]. On top of that, the rate of under-reporting may vary depending on specific drugs, the type of event and the severity of an event [53], which further biases the results. A true estimate of the actual incidence and prevalence of MEs is not possible as our information is based on ICSRs only. The lack of data on age and sex in many ICSRs inhibits the use of all ICSRs for subgroup analyses, and therefore the disproportionality analyses on the subgroups may have produced biased SDRs. At the same time, the abundance of data provides a challenge in prioritising, selecting, sorting and displaying the data. An analysis based on country level is of interest and could provide interesting findings in future research. Furthermore, the ROR and its 95% CI should be carefully interpreted, especially if a low number of ICSRs contributes to the SDR. While it is beyond the scope of this paper, the unavailability of easily accessible ICSR narratives hinders further in-depth investigation of newly found SDRs for MEs. Because of the study design, we only analysed the MEs presented as PTs in the ICSRs, while based on the raw data, there might have been more MEs present. For example, identification of contra-indications, wrong dosing and inappropriate polytherapy would have required additional analysis on more data such as (assessing appropriate) age, presence of pregnancy and use of concomitant medications. While most ICSRs mention one suspect drug, 20% of the ICSRs mention two or more suspect drugs. It is important to note that not all the suspect drugs within a given report might be associated with multiple MEs in the same report; however, the disproportionality analysis mitigates this issue. Additionally, a unique ICSR may contain multiple PTs describing the same ME, which might contribute to overestimation of a reported MEs in some cases. Some drugs, for which multiple ATC codes were available, could not be mapped to a single ATC code because of a lack of information, which might further contribute to potentially biased results.

5 Conclusions

This research provides an overview of MEs and SDRs from the EudraVigilance database. A large number of SDRs were identified, part of which are not described in the literature and may prompt further investigation by stakeholders. For drugs with the most reported MEs, the most conspicuous

SDRs were identified in specific age groups, especially for vaccines, paracetamol, oxycodone, rivaroxaban and risperidone. New SDRs were identified for DMARDs for which the ME "Contraindicated product administered" was most often reported.

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Declarations

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Conflicts of Interest/Competing Interests The research group from the Department of Medical Informatics received unconditional research grants from Chiesi, UCB, Amgen, Johnson and Johnson, and the European Medicines Agency, none of which relates to the content of this article.

Ethics Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Availability of Data and Material An agreement has been made with the EMA not to transfer or dispose of the ICSR data set Level 2A for which access is provided under the condition of the Confidentiality Undertaking to any third party. It is agreed not to permit any third party to access, study, analyse, refer to or otherwise use the data or permit any party to reproduce any ICSR data. A publicly available version of this database is available here: https://www.adrreports.eu/en/index.html. Further inquiries can be directed to the corresponding author.

Code Availability Not applicable.

Authors' Contributions VP: conceptualisation, formal analysis, visualisation, writing (original draft), writing (review and editing), methodology, visualisation. JK: software, writing (review and editing). EV: software, writing (review and editing). MdW: software, writing (review and editing). PR: supervision, writing (review and editing). KV: conceptualisation, supervision, writing (review and editing). All authors read and approved the final version.

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